

MakeCircleMatchingFilter

Create a function **MakeCircleMatchingFilter**, which you will use in your final project script. It takes as input the **diameter** for the circle as filter. It outputs the [WxW] matrix **filter**, which is a binary mask that contains, as foreground, a circle with width **diameter** centered in the r **filter** serves as a matching filter for circular shapes with approximately the same diameter. The function also outputs **xc** and **yc**, the coor and 3rd optional outputs.

The steps to perform in the function are as follows:

- Initialize **filter** as a background (zeros) matrix.
- Find the coordinates of the center of the **filter**, **xc**, and **yc**.
- Use nested for loops to loop across every row and column of the filter matrix. For each pixel, determine whether the pixel's row-c radius relative to the center of the filter. If so, should be labelled as foreground with a 1.

When running the example code below, your function should result in output identically to this:

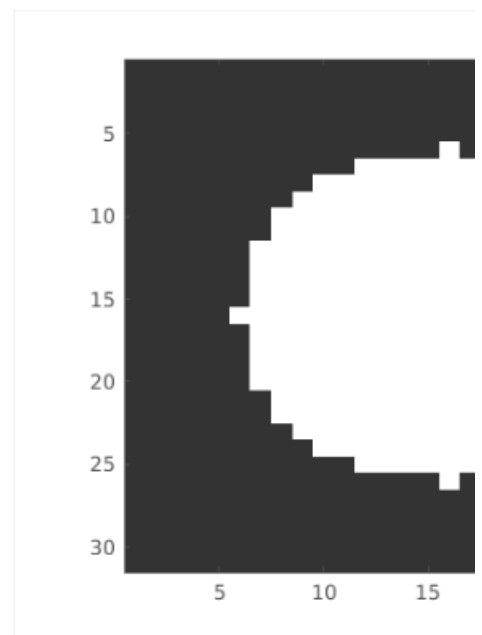
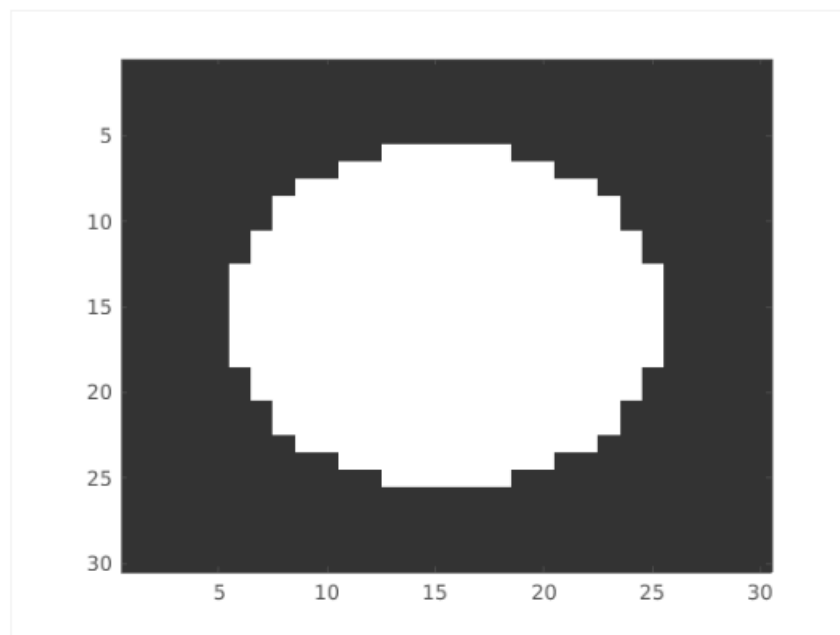
Output

```
xc =  
  
    15.5000
```

```
yc =  
  
    15.5000
```

```
xc =  
  
    16
```

```
yc =  
  
    16
```



Function ?

Save

Reset

MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 function [filter,xc,yc] = MakeCircleMatchingFilter(diameter,W)
2 % initialize filter
3 filter = zeros(W, W);
4 % define coordinates for the center of the WxW filter
5 xc = (W+1)/2;
6 yc = (W+1)/2;
7 % Use double-for loops to check if each pixel lies in the foreground of the circle
8 r = diameter / 2;
9 for i=1:W
10     for j=1:W
11         if (i-xc)^2 + (j-yc)^2 <= r^2
12             filter(i,j) = 1;
13         end
14     end
15 end
16
17 end
18
```

Code to call your function ?

 Reset

```
1 diameter = 20;
2 W = 30;
3 [filter, xc, yc] = MakeCircleMatchingFilter(diameter,W);
4 imagesc(filter); colormap(gray)
5 xc % should be 15.5
6 yc % should be 15.5
7
8 diameter = 20;
9 W = 31;
10 [filter, xc, yc] = MakeCircleMatchingFilter(diameter,W);
11 figure; imagesc(filter); colormap(gray);
12 xc % should be 16
13 yc % should be 16
```

 Run Function

Assessment: All Tests Passed

Submit ?

 Is xc correct? Is yc correct? Is filter correct?